Equity Pesalink V2 Outbound

Business Service

V 1.0.2

Revision History

Date	Version	Author	Comments
01.08.2021	v1.0.0	Kennedy N. Gatimu	Initial document creation
02.08.2021	v1.0.1	Ronald K. Cheruiyot	First revision
06.08.2021	v1.0.2	Kennedy N. Gatimu	Second revision - added signature to transaction and status check API payloads

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1.0 Introduction

The Pesalink Outbound Business Service is a client-facing gateway system that facilitates the transfer of funds to external banks and fintechs. This document presents information about the APIs exposed on the system and through which Equity bank channels such as STK and USSD can initiate name check, credit transfer and transaction status check requests. It also contains information about how the client channels should implement digital signature functionality in order to ensure that the exchange of messages between the channel and the gateway is secure.

2.0 API Reference

2.1.0 Name Check API

The name check API will return the verified name of the receiver. Based on the response, the customer can decide to proceed with or cancel the credit transfer. This API records all the information that will be used to complete the subsequent transaction/credit transfer and transaction status check calls such that in these two requests, only the original request ID is provided.

Note: For destination banks and fintechs designated as connecting through the converter as shown in appendix 1 & 2, the name check request will not return a verified name because the implementation is not yet done on their side. Hence, credit transfer can be initiated even if the name check fails.

Table 2.1: Name check API endpoint description

Method	POST
Endpoint URL	/v1/verification-request
Headers	Content-Type:application/json

Name Check API Request Fields

Table 2.2: Name check API request fields

Field	**Description**	**Type**	**Mandatory**
requestId	Unique identifier for the message	String	Y
channelId	Unique identifier for the originator channel[1]	String	Y
sourceSystemId	Unique identifier for the source system ID	String	Y
callbackURL	URL that will be used to send a call back request	String	Y
sourceTranType	Transaction type	String	Y
sourceAccount	Bank account number of the sender	String	Y
sourceCountryCode	Country code of the sender bank account (Use ISO 3166-1 alpha-3 standard)	String	Y
sourceCurrency	Currency code of the sender bank account (Use ISO 4217 standard)	String	Y

[1] Check section 2.3.0 for default allocated IDs

senderName	Name of the sender	String	Y
senderInstitutionCode	PIC code of the sender, default value is ‘0068’	String	Y
senderPhone	Mobile phone number of the sender	String	N
receiverAccount	Bank account number of the receiver	String	Y
receiverInstitutionCode	PIC code of the receiver	String	Y
receiverCurrency	Currency code of the receiver bank account (Use ISO 4217 standard)	String	Y
amount	Amount to be sent	String	Y
feeAmount	Amount of transaction fees to be applied as requested by the source channel, default value is ‘0’	String	Y
paymentReason	Reason for payment, e.g: ‘School fees’	String	Y
narration	Narration to be sent with the payment, for example the registration number of student	String	Y
signature	Digital signature generated from the request payload	String	Y

Name Check API Response Fields

Table 2.3: Name check API response fields

Field	Description	Type
requestId	Unique identifier for the message	String
endToEndId	Unique identifier for the transaction	String
channelId	Unique identifier for the originator channel	String
sourceSystemId	Unique identifier for the source system ID	String
receiverAccount	Bank account number of the receiver	String
receiverInstitutionCode	PIC code of the receiver	String
receiverName	Name of the receiver	String
status	Status to show whether the message was	String

	processed successfully	
statusCode	Status code	String
statusDesc	Status description	String

Sample Name Check Request


```
{
  "requestId":"EQUITY000000000105",
  "channelId":"STK",
  "sourceSystemId":"EQUITEL_STK",
  "callbackURL":"http://localhost:8085/v1/callback/channel",
  "sourceTranType":"1",
  "sourceAccount":"1001008010940",
  "sourceCountryCode":"54",
  "sourceCurrency":"KES",
  "senderName":"Jane Doe",
  "senderInstitutionCode":"0068",
  "senderPhone":"+254-720000000",
  "receiverAccount":"0170197176640",
  "receiverInstitutionCode":"0012",
  "receiverCurrency":"KES",
  "amount":"500",
  "feeAmount":"0",
  "paymentReason":"Payments",
  "narration":"Pesalink Transfer",
"signature":"hBJ9FAEzPMaZ37V/iOSTLqxmJfwEBkeC9+cskTmnIf79kxsgbpSyijNAZ3XXxm9kHuN5qPt
qJ9tijcbUra9l1YCter8nyLPJbB25FVOmBpU2+RNCoPQqmUWZETwtpl5lVIxfSuA3Uk+XB9s1XBqn43bkFZM
qPbGGascKXB41wvFAmT3gEhpHfy4OFhIbJa2qS2JBcdX2ZWHBHpeLdZ8iLAUTSln+CwXTv7749el/dWXxDhH
5dSDYGehDrem0QLdFSwUYa5vTmajuBd4FdWQQkWZjLUYcg1M9NRVuWhHTW9YBR0bo+Nb8P8VOPUjMtHL7Odu
QLdebWVrr7TDC7zf9Kg=="
}
```

Sample Name Check Response - Success


```
{
  "requestId":"EQUITY000000000105",
  "endToEndId":"123456789",
  "channelId":"STK",
  "sourceSystemId":"EQUITEL_STK",
  "receiverAccount":"0170197176640",
  "receiverInstitutionCode":"0012",
  "receiverName":"John Doe",
  "status":"success",
  "statusCode":"00",
  "statusDesc":"Request Processed Successfully"
}
```


Sample Name Check Response - Failure


```
{
  "requestId":"EQUITY000000000105",
  "channelId":"STK",
  "sourceSystemId":"EQUITEL_STK",
  "receiverAccount":"1145756190",
  "receiverInstitutionCode":"0001",
  "status":"Failed",
  "statusCode":"01",
  "statusDesc":"Request Failed. Internal Exception Occurred"
}
```


Name Check Response HTTP Status Codes

Table 2.4: Name check API response HTTP status codes

HTTP Status Code	HTTP Status Code Description
200	Ok
202	Accepted
4xx	Request was not accepted

2.2.0 Transaction API

The name transaction/credit transfer API will be used to initiate a transaction based on the information passed during the name check request. It only takes the original request ID passed during the name check stage.

Table 2.5: Transaction API endpoint description

Method	POST
Endpoint URL	/api/v1/credit-transfer
Headers	Content-Type:application/json

Transaction API Request Fields

Table 2.6: Transaction API request fields

Field	Description	Type	Mandatory
requestId	Request ID of the original name check request	String	Y
channelId	Unique identifier for the originator channel[2]	String	Y
sourceSystemId	Unique identifier for the source system ID	String	Y
signature	Digital signature generated from the request payload	String	Y

Transaction API Response Fields

Table 2.7: Transaction API response fields

Field	Description	Type
requestId	Request ID of the original name check request	String
status	Status to show whether the message was processed successfully	String
statusCode	Status code	String
statusDesc	Status description	String

[2] Check section 2.3.0 for default allocated IDs

Sample Transaction Request

```
{
  "requestId":"EQUITY000000000105",
  "channelId":"STK",
  "sourceSystemId":"EQUITEL_STK",

"signature":"d56Ad+4beRHK/7Eu5AxcbYtgAkl4SyWieRTQJvCzev3RDJ0HDYB4pSQOLhKXVhMXDY4EnZp
ZgAEQSWlJ3hYgA8WYpuOkVmlKwOSg5fQKMwo+6ydWEllhzMrsVHJrouze5KIh5aFTR8sXMvGA3bUeOvpzz4T
ZvdzdLH6CYe6Oe2sB+h/2Ukm21VPbOqjzizwtWETptppVagSwq+oV0rRCNX8hDbFY5W5maiuqrKk3l4fbY1T
KEYASzH1Km1P/tUrlj48WFOhT771OvcdZb+9jz6AQkP8+gPgk/OgKRfujaNgEdM67b3tcZpJTEuy6W+Wwieq
jL/G3udBSs95TwqTAIBw=="
}
```

Sample Transaction Response - Success

```
{
  "requestId":"EQUITY000000000105",
  "status":"Success",
  "statusCode":"00",
  "statusDesc":"Request accepted successfully"
}
```

Sample Transaction Response - Failure

```
{
  "requestId":"EQUITY000000000105",
  "status":"Failed",
  "statusCode":"01",
  "statusDesc":"Request failed"
}
```

Transaction Response HTTP Status Codes

Table 2.8: Transaction API response HTTP status codes

HTTP Status Code	HTTP Status Code Description
200	Ok
202	Accepted
4xx	Request was not accepted

2.2.0 Transaction Status Check API

The name transaction status check API will be used to check the status of the transaction initiated during the credit transfer stage based on the information passed during the name check request. It only takes the original request ID passed during the name check stage.

Table 2.9: Transaction status check API endpoint description

Method	POST
Endpoint URL	/v1/payment-status-request
Headers	Content-Type:application/json

Transaction Status Check API Request Fields

Table 3.0: Transaction status check API request fields

Field	Description	Type	Mandatory
requestId	Request ID of the original name check request	String	Y
channelId	Unique identifier for the originator channel[3]	String	Y
sourceSystemId	Unique identifier for the source system ID	String	Y
signature	Digital signature generated from the request payload	String	Y

Transaction Status Check API Response Fields

Table 3.1: Transaction status check API response fields

Field	Description	Type
requestId	Unique identifier for the message	String
endToEndId	Unique identifier for the transaction	String
channelId	Unique identifier for the originator channel	String
sourceSystemId	Unique identifier for the source system ID	String
callbackURL	URL that will be used to send a call back request	String
sourceTranType	Transaction type	String

[3] Check section 2.3.0 for default allocated IDs

sourceAccount	Bank account number of the sender	String
sourceCountryCode	Country code of the sender bank account (ISO 3166-1 alpha-3 standard)	String
sourceCurrency	Currency code of the sender bank account (ISO 4217 standard)	String
senderName	Name of the sender	String
senderInstitutionCode	PIC code of the sender, default value is ‘0068’	String
senderPhone	Mobile phone number of the sender	String
receiverAccount	Bank account number of the receiver	String
receiverInstitutionCode	PIC code of the receiver	String
receiverCurrency	Currency code of the receiver bank account (ISO 4217 standard)	String
amount	Amount to be sent	String
feeAmount	Amount of transaction fees to be applied as requested by the source channel, default value is ‘0’	String
paymentReason	Reason for payment, e.g: ‘School fees’	String
narration	Narration captured in the original name check request	String
dateAdded	Date transaction was registered	String
transactionCount	Number of transactions	String
status	Status to show whether the message was processed successfully	String
statusCode	Status code	String
statusDesc	Status description	String
statusMessage	Any other information associated with the transaction status	String

Sample Transaction Status Check Request

```
{
  "requestId":"EQUITY000000000105",
  "channelId":"STK",
  "sourceSystemId":"EQUITEL_STK",

"signature":"d56Ad+4beRHK/7Eu5AxcbYtgAkl4SyWieRTQJvCzev3RDJ0HDYB4pSQOLhKXVhMXDY4EnZp
ZgAEQSWlJ3hYgA8WYpuOkVmlKwOSg5fQKMwo+6ydWEllhzMrsVHJrouze5KIh5aFTR8sXMvGA3bUeOvpzz4T
ZvdzdLH6CYe6Oe2sB+h/2Ukm21VPbOqjzizwtWETptppVagSwq+oV0rRCNX8hDbFY5W5maiuqrKk3l4fbY1T
KEYASzH1Km1P/tUrlj48WFOhT77lOvcdZb+9jz6AQkP8+gPgk/OgKRfujaNgEdM67b3tcZpJTEuy6W+Wwieq
jL/G3udBSs95TwqTAIBw=="
}
```

Sample Transaction Status Check Response - Success

```
{
  "requestId":"EQUITY000000000105",
  "endToEndId":"0068000120210726113234bac05177",
  "channelId":"STK",
  "sourceSystemId":"EQUITEL_STK",
  "callbackURL":"http://localhost:8085/v1/callback/channel",
  "sourceTranType":"account",
  "sourceAccount":"0170197176640",
  "sourceCountryCode":"54",
  "sourceCurrency":"KES",
  "senderName":"Jane Doe",
  "senderInstitutionCode":"0068",
  "senderPhone":"+254-720000000",
  "receiverAccount":"1145756190",
  "receiverInstitutionCode":"0001",
  "receiverCurrency":"KES",
  "amount":"500",
  "feeAmount":"0",
  "paymentReason":"Payments",
  "narration":"Pesalinks Transfer",
  "dateAdded":"2021-07-26T11:31:57.275Z",
  "transactionCount":"0",
  "status":"Success",
  "statusCode":"00",
  "statusDesc":"Request processed successfully",
```

```
  "statusMessage":"Request was processed successfully by IPS"
}
```


Sample Transaction Status Check Response - Failure


```
{
  "requestId":"EQUITY000000000105",
  "endToEndId":"0068000120210726113234bac05177",
  "channelId":"STK",
  "sourceSystemId":"EQUITEL_STK",
  "callbackURL":"http://localhost:8085/v1/callback/channel",
  "sourceTranType":"account",
  "sourceAccount":"01701971766487",
  "sourceCountryCode":"54",
  "sourceCurrency":"KES",
  "senderName":"Jane Doe",
  "senderInstitutionCode":"0068",
  "senderPhone":"+254-720000000",
  "receiverAccount":"1145756190",
  "receiverInstitutionCode":"0001",
  "receiverCurrency":"KES",
  "amount":"500",
  "feeAmount":"0",
  "paymentReason":"Payments",
  "narration":"Pesalinks Transfer",
  "dateAdded":"2021-07-26T11:31:57.275Z",
  "transactionCount":"0",
  "status":"Failed",
  "statusCode":"01",
  "statusDesc":"Request Failed.",
  "statusMessage":"Request Failed."
}
```

Transaction Response HTTP Status Codes

Table 2.8: Transaction status check API response HTTP status codes

HTTP Status Code	HTTP Status Code Description
200	Ok
202	Accepted
4xx	Request was not accepted

2.3.0 Channel and Source System IDs

The following are the channel & source system IDs allocated for the various channels available. Whenever a channel ID is passed in a payload, the correct corresponding source system ID should also be supplied.

Table 2.9: Channel and source system IDs

Channel ID	Source System ID
STK	EQUITEL_STK
APP	EQUITY_APP
WEB	EQUITY_WEB
USSD	EQUITY_USSD
EAZAPP	EAZZY_APP
EAZNET	EAZZY_NET
EAZBIZ	EAZZY_BIZ
EVA	CHATBOT_EVA
OTC	BRANCH

3.0 Digital Signature

For the purpose of ensuring data integrity and security, every channel will be required to generate a digital certificate and share the same with the maintainer of this API. The certificate will be used to verify the signature contained in each payload.

3.0.1 Generating a Cryptographic Key Pair and CSR

One can generate a 2048-bit RSA key using the following Openssl command:

```
openssl req -new -newkey rsa:2048 -nodes -keyout demo.key -out demo.csr
```

Enter the required details as below:

```
Country Name (2 letter code): KE

State or Province Name (full name): Nairobi

Locality Name: Nairobi

Organization Name: Equity

Organizational Unit Name: pesalink

Common Name: equitybankgroup.co.ke

Email Address: info@equitybankgroup.co.ke

Challenge password: demo
```

3.0.2 Generating a Digital Certificate

Once the CSR has been generated, one is required to acquire a valid certificate signed by a recognized CA, but for the purpose of testing, one can generate a self-signed certificate in *.PKCS12* format using the Openssl commands below.

1. Generate a self-signed certificate in .crt format:

```
openssl x509 -req -days 365 -in demo.csr -signkey demo.key -sha256 -out demo.crt
```

2. Convert the .crt certificate to .p12 format

```
openssl pkcs12 -export -in demo.crt -inkey demo.key -out demo.p12 -name demo
```

3. Share the *.PKCS12* certificate with the maintainer of this API so that it can be configured in the system

3.0.2 Generating the Signature

Each request should have a digital signature generated as described below:

1. Generate the clear text:
   Concatenate the values in each request field shown below to generate the clear text. The clear text value will be hashed.
   a. For Account Verification API, generate the clear text as:
      clearText = requestId
      + channelId
      + sourceSystemId
      + sourceAccount
      + sourceCurrency
      + senderName
      + senderInstitutionCode
      + receiverAccount
      + receiverInstitutionCode
      + receiverCurrency
      + amount
      + feeAmount

   b. For Credit Transfer API, generate the clear text as:
      clearText = requestId

   c. For Status Check API, generate the clear text as:
      clearText = requestId

2. Sign the clear text:
   Sign the request using the clear text generated in step 1, the RSA private key generated in section 3.0.2 and by using the SHA-256 hashing algorithm. At this point the output should be a byte array.
   In summary, the signature scheme should be SHA-256 with RSA

The figure below shows an example of how the signature can be generated using Java although the same can be achieved using any programming language

Fig. 1: Example of how to generate a digital signature using Java

```java
import java.security.PrivateKey;
import java.security.Signature;
import java.util.Base64;


public class SignMsg {

   private static final String UTF_8 = "UTF-8";
   private static final String ALGORITHM = "SHA256WithRSA";


   public String signMessage(String theXsumFields, PrivateKey privatekey) {
       try {
           byte[] messageBytes = theXsumFields.getBytes(UTF_8);
           Signature signatureInstance = Signature.getInstance(ALGORITHM);
           signatureInstance.initSign(privatekey);
           signatureInstance.update(messageBytes);
           byte[] digitalSignature = signatureInstance.sign();
           String signature = Base64.getEncoder().encodeToString(digitalSignature);
           return signature;
       } catch (Exception ex) {
           ex.printStackTrace();
           return null;
       }

   }
}
```

Appendix 1: List of Participant Identification Codes(PIC) For Banks

BANK NAME	PARTICIPANT IDENTIFICATION CODE(PIC)	CONNECTING THROUGH
KCB	0001	DIRECT
Stanchart	0002	CONVERTOR
ABSA	0003	DIRECT
Bank of India	0005	DIRECT
Bank of Baroda	0006	DIRECT
NCBA	0007	DIRECT
Prime Bank	0010	DIRECT
Coop Bank	0011	DIRECT
NBK	0012	DIRECT
M-Oriental	0014	DIRECT
Citi Bank	0016	CONVERTOR
Habib Bank AG Zurich	0017	DIRECT
Middle East Bank	0018	CONVERTOR
Bank of Africa	0019	DIRECT
Consolidated	0023	DIRECT
Credit Bank	0025	DIRECT
Access Bank	0026	DIRECT
Stanbic Bank	0031	DIRECT
ABC Bank	0035	CONVERTOR
Eco Bank	0043	DIRECT
SPIRE Bank	0049	DIRECT
Paramount	0050	DIRECT
Kingdom Bank	0051	CONVERTOR
Gt Bank	0053	CONVERTOR
Victoria Bank	0054	CONVERTOR
Guardian Bank	0055	CONVERTOR
I&M Bank	0057	DIRECT
Development Bank	0059	DIRECT
SBM	0060	DIRECT
Housing finance	0061	CONVERTOR

DTB	0063	DIRECT
Mayfair Bank	0065	DIRECT
Sidian Bank	0066	CONVERTOR
Equity Bank	0068	DIRECT
Family Bank	0070	CONVERTOR
Gulf African Bank	0072	CONVERTOR
First Community Bank	0074	CONVERTOR
DIB Bank	0075	CONVERTOR
UBA	0076	DIRECT
KWFT	0078	CONVERTOR
Faulu Bank	0079	DIRECT
Post Bank	0099	DIRECT
IPS (Instant Payment Switch)	9999	

Appendix 2: List of Participant Identification Codes(PIC) For Fintechs

FINTECH NAME	PARTICIPANT IDENTIFICATION CODE(PIC)	CONNECTING THROUGH	TEMPORARY PIC ASSIGNED
Telkom Kenya	0097	DIRECT	0097
Airtel Kenya	TO BE CONFIRMED	DIRECT	
Eclectics	0096	DIRECT	0096
MFS	0095	IPSL PAYMENT GATEWAY	0095
Kocela	0094	IPSL PAYMENT GATEWAY	0094
KCBbulk	0093	IPSL PAYMENT GATEWAY	0093
iPay	0092	IPSL PAYMENT GATEWAY	0092
Flutterwave	0091	IPSL PAYMENT GATEWAY	0091
Cellulant	0090	DIRECT	0090
Stima Sacco	0089	IPSL PAYMENT GATEWAY	0089
Interswitch	0088	IPSL PAYMENT GATEWAY	0088
Craft Silicon	TO BE CONFIRMED	TO BE CONFIRMED	
Kenswitch	TO BE CONFIRMED	DIRECT	
Pesapal	0087	IPSL PAYMENT GATEWAY	0087
Pesaswap	TO BE CONFIRMED	TO BE CONFIRMED	
MyKeekapu	TO BE CONFIRMED	TO BE CONFIRMED	
Sybyl Kenya Ltd	TO BE CONFIRMED	TO BE CONFIRMED	
Power	TO BE CONFIRMED	TO BE CONFIRMED	
Sevi	TO BE CONFIRMED	TO BE CONFIRMED	
Virtual City	TO BE CONFIRMED	TO BE CONFIRMED	
KAPS	TO BE CONFIRMED	TO BE CONFIRMED	
Pendo Media	TO BE CONFIRMED	TO BE CONFIRMED	